CSCI 5561 Homework 1

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1 Introduction

In this project, I build a panoramic image stitcher from scratch. Given two overlapping photographs, my system detects and matches local features, estimates the geometric transformation relating the images, and warps them into a common coordinate frame to produce a panorama. I also implement and analyze additional features.

2 Image Capture and Feature Extraction

I capture two sets of images. One set consists of an atrium in which I rotate between pictures. A homography is appropriate for this movement since rotation from a single point does not produce any depth dependent parallax. This allows a homography to align the scene even when there is depth variation. The other set consists of a wall of paintings where I translate to the right. A homography is appropriate for this set despite camera translation since the subject forms an approximately flat scene. This was all shot on an iPhone 17 and I used SIFT and set nfeatures to be 3,000. With this, my Atrium 1 and Atrium 2 images had 3000 keypoints, my Paintings 1 and Paintings 3 images had 1460 and 1498 respectively.

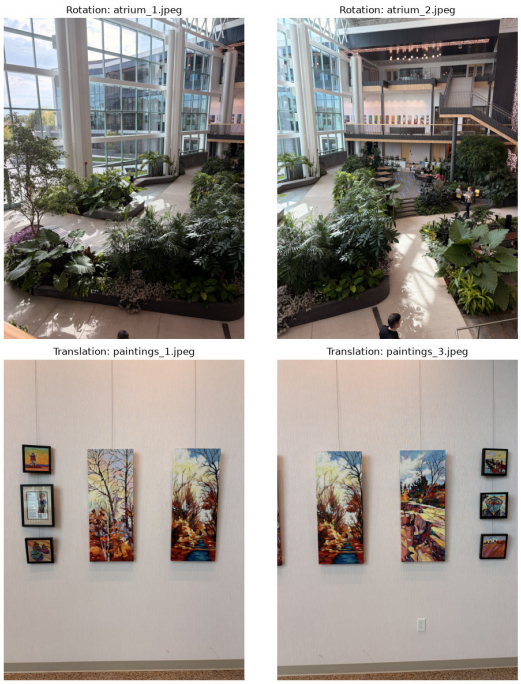


Figure 1: Two Image Sets

3 Descriptor Matching

For descriptor matching, I selected a threshold of 0.75. This served as a baseline and provided good results. We compute the euclidean distance between each descriptor and find the two nearest neighbors. After this we accept matches that pass the Lowe's ratio test with the threshold of 0.75. For the Atrium set, there were 3,000 candidate descriptors which were narrowed down to 354 after the ratio test. For the paintings set there were 1,460 candidate descriptors which narrowed down to 387.



Figure 2: 100 Strongest SIFT Keypoints from Atrium 1



Figure 3: Before and After Lowe's Ratio Test

4 Homography Estimation

I estimate a homography using at least 4 corresponding keypoint coordinates across two images. This is done first through normalizing source and destination points. I then build a DLT system with two equations per correspondence. I then solve this using the furthest right singular vector and then undo the normalization.

For validation, I used the known homography:

$$H_{\text{known}} = \begin{bmatrix} 1.1 & 0.1 & 30 \\ -0.05 & 0.9 & 20 \\ 0.0005 & -0.0003 & 1 \end{bmatrix}$$

The two tests used 4 and 10 correspondences respectively. The 4 point test used rectangular corners so no three were collinear. The ten point test added 6 points distributed through the rectangle. For each test, I generated destination points by applying the known homography to the source points. I then estimated a homography from these correspondences and measured the mean Euclidean distance between its mapped source points and the destination points. This compares the transformations through their mapped points rather than their matrix entries. Below are the mean reprojection errors for the two tests.

Correspondences	Mean reprojection error (pixels)
4	1.998×10^{-14}
10	2.986×10^{-14}

Table 1: Mean Reprojection Error for Correspondences.

5 Basic Robust Estimation with RANSAC

I used an inlier threshold of 3 pixels along with a fixed iteration budget of 2,000. These both served as a baseline. 2,000 iterations were used as a conservative fixed baseline for finding a useful sample. We use 3 pixels as a balance to account for noise but also remove actual outliers. For RANSAC, I take random samples containing 4 correspondences per trial to estimate homographies. I also discard estimated homographies that have nonfinite entries or rank below 3. After 2,000 iterations the homography with the highest consensus is used and I refit a new homography with that inlier set. For the atrium set, there were 354 total matches. Of those, 210 were inliers which resulted in an inlier ratio of 59.3%. The mean inlier reprojection error before the final refit was 1.357 pixels and 1.123 pixels

after. For the paintings set, there were 387 total matches with 363 inliers which resulted in an inlier ratio of 93.8%. The mean inlier reprojection error before the final refit was 0.656 pixels and 0.347 pixels after. Reprojection errors used same inlier sets respectively. The refits for both of these sets were a result of fitting a new homography on all inliers instead of just the 4 random correspondences.



Figure 4: Visualization of RANSAC Inliers and Outliers

6 Warping and Compositing

For warping and compositing, I used atrium_2 and paintings_3 as references. I refer to these references as B and other image as A. I then transform A's corners to B using a homography before combining corners to compute canvas bounds. I then apply a translation matrix T to account for negative coordinates. The transformations for warping are then:

$$G_A = TH_{A\to B}, \qquad G_B = T.$$

To sample the source images, for every canvas pixel we take G inverse to find the source image coordinates and divide by the homogeneous coordinate. I also used map_coordinates with order=1 for bilinear interpolation. I track a separate validity mask for each warped image. A canvas pixel is valid when its inverse mapped source coordinates lie within the source image bounds. For overwrite compositing, I place image A first, then overwrite it with image B wherever B's mask is valid.

For synthetic testing, the known transformations are

$$H_{\text{translation}} = \begin{bmatrix} 1 & 0 & 30 \\ 0 & 1 & 20 \\ 0 & 0 & 1 \end{bmatrix}, H_{\text{scale}} = \begin{bmatrix} 1.5 & 0 & 0 \\ 0 & 1.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

which are a translation to the right and down and a scale by 1.5.

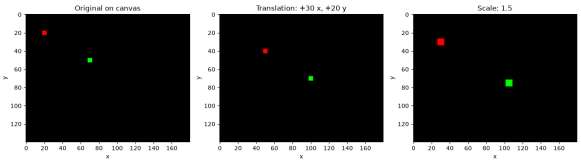


Figure 5: Warp Test

The identifiable image locations move as expected with the red center moving from (20,20) to (50,40) under translation and (30,30) under scaling. The green center moves from (70,50) to (100,70) under translation and (105,75) under scaling.

For the captured sets, the atrium set had output canvas dimensions of 1357 x 1319 pixels. The paintings set has output dimensions of 974 x 826 pixels. The reference picture for both cases is the right image.

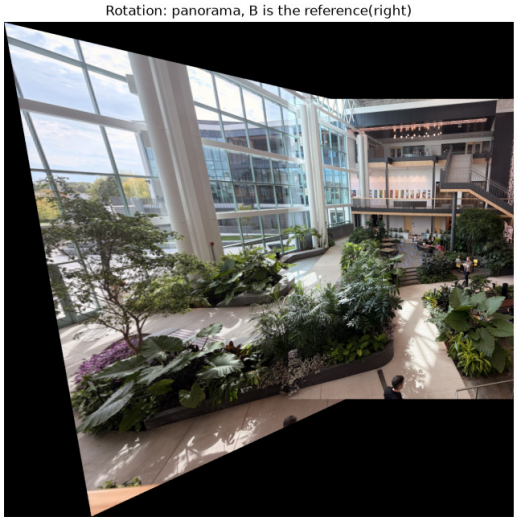


Figure 6: Panorama of Atrium Set, Reference Image is Right Side

7 Feather Blending

I decided to replace overwrite compositing with a feather blend whose weight changes across the overlap. Each warped image receives weights based on distance from boundary. Interior pixels have more weight while boundary pixels have less weight.

With the addition of feather blending, the seam is much less noticeable, making the panorama look much more cohesive. The abrupt brightness change is gone and the transition is smoother. The blended image is a bit blurrier compared to overwrite compositing. This is apparent in the tree in figure 10 where the details of the tree look softer. This can

Pair	Keypoints A / B	Candidate descriptors	Ratio test matches	Inliers	Inlier ratio	Mean error (px) before / after refit	Canvas (px) width × height
Rotation	3000 / 3000	3000	354	210	59.3%	1.357 / 1.123	1357 × 1319
Translation	1460 / 1498	1460	387	363	93.8%	0.656 / 0.347	974 × 826

Table 2: Results for both image pairs. Image A is the first image and image B is the reference.



Figure 7: Panorama of Paintings Set, Reference Image is Right Side

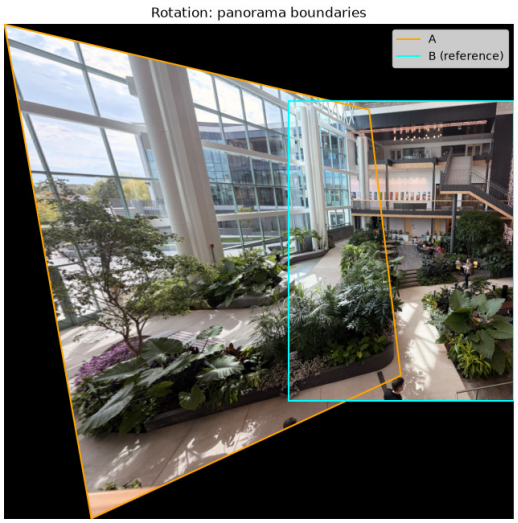


Figure 8: Visualization of Boundaries in Atrium Panorama, Right Image is Reference



Figure 9: Overwrite vs Feather Blending for Atrium Set



Figure 10: Close Up of Feather Blending Comparison

be attributed to feather blending averaging slightly misaligned views of the tree across the overlap.

8 Confidence-Based RANSAC Stopping

In this section, I implement confidence based RANSAC stopping. With this, I dynamically update the RANSAC iteration count using a target confidence and best inlier fraction found so far. I start with a max budget of 2,000 iterations and estimate an inlier fraction w whenever a larger consensus is found. I then recalculate the required iterations using my confidence target. I chose a confidence target of 0.99 with these formulas:

$$w = \frac{I_{\text{best}}}{M}$$

$$N_{\text{required}} = \left\lceil \frac{\log(1-p)}{\log(1-w^4)} \right\rceil .$$

Pair	Fixed iterations	Adaptive iterations	Fixed inliers	Adaptive inliers
Rotation	2000	75	210	175
Translation	2000	4	363	361

Table 3: Fixed VS Adaptive RANSAC. Adaptive stopping uses a 99% confidence target

As can be seen in table 3, this adaptive RANSAC greatly reduces the number of iterations required. For translation, almost no inliers were lost while for rotation about 17% of inliers were lost.

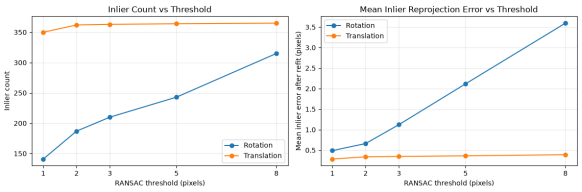


Figure 11: Plot of Inlier Count and Mean Inlier Reprojection Error

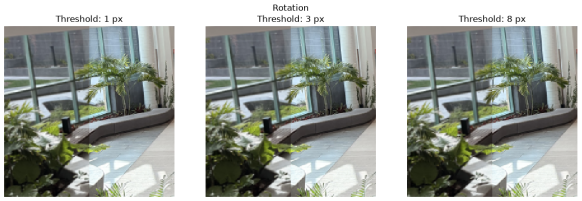


Figure 12: Rotation Panorama Crops Across Thresholds

9 RANSAC Parameter Analysis

For this section, I test a variety of inlier thresholds with 1, 2, 3, 5, and 8 pixels. I kept the same matches and 2,000 iteration budget.

The rotation pair was more sensitive to the threshold. Increasing it from 1 to 8 pixels increased the inlier count from 141 to 315, while the mean error increased from 0.489 to 3.596 pixels. At the baseline threshold of 3 pixels, 210 matches were retained with a mean error of 1.123 pixels. At the highest threshold, there appears to be a visible alignment error suggesting that it mainly admits worse matches.

The translation pair changed much less. Increasing the threshold from 1 to 8 pixels increased the inlier count from 350 to 365, with mean error increasing from 0.285 to 0.390 pixels. Above 3 pixels, only two additional matches were accepted. The crops appeared nearly identical, suggesting little benefit from a bigger threshold.

10 Interpolation Comparison

I compared nearest neighbor and bilinear interpolation using the rotation panorama. Both versions used the same homography, canvas bounds, and overwrite compositing, so only the sampling strategy changed. I used map_coordinates with interpolation orders 0 and 1 respectively.

Nearest neighbor interpolation produced more visible step edges because it selects the closest source pixel. Bilinear interpolation produced smoother and softer boundaries by taking a weighted average of four neighboring pixels. The difference image shows the mean absolute RGB difference. Differences are concentrated around the edges, where neighboring pixels have different intensities.



Figure 13: Translation Panorama Crops Across Thresholds

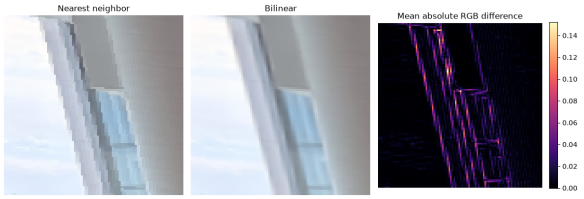


Figure 14: Crop Comparing Nearest Neighbor and Bilinear with Mean RGB Difference

11 Least Squares versus RANSAC

I fitted a homography with every tentative match and compared it with the RANSAC result. The all match homography produced large distortion in both sets. In rotation, the windows, plants, and floor were strongly stretched. In translation, the left painting region was enlarged and did not align with the reference image. An inlier visualization can be seen at Figure 4.



Figure 15: Comparing Panorama with All Matches vs RANSAC Rotation

12 Detector and Matcher Comparison

I compared SIFT, ORB, and AKAZE on both image pairs. SIFT and ORB used a maximum of 3,000 features, while AKAZE used its default settings. SIFT used my custom matcher, ORB and AKAZE



Figure 16: Comparing Panorama with All Matches vs RANSAC Translation

used OpenCV's Hamming distance matcher. All methods used a ratio threshold of 0.75 and my homography estimation, RANSAC, and warping code. RANSAC used a 3 pixel threshold and 2,000 iterations.

Pair	Method	Keypoints A/B	Matches	Inliers	Inlier ratio	Runtime (s)
Rotation	SIFT	3000/3000	354	210	59.3%	1.031
Rotation	ORB	3000/3000	193	151	78.2%	0.694
Rotation	AKAZE	2433/2492	307	200	65.1%	0.291
Translation	SIFT	1460/1498	387	363	93.8%	0.352
Translation	ORB	2981/3000	584	551	94.3%	0.238
Translation	AKAZE	1158/1075	304	284	93.4%	0.266

Table 4: Detector and Matcher Comparison.

I used the rotation pair as the more challenging comparison because many matches were geometrically inconsistent with only 59.3% of the SIFT matches being retained by RANSAC.

Runtime includes feature extraction, matching, and RANSAC. The comparison figure shows sampled matches, with green inliers and red outliers, alongside panoramas for rotation. All panoramas use the same canvas, second image as reference, bilinear interpolation, and overwrite compositing. ORB produced fewer rotation matches but the highest inlier ratio, while AKAZE obtained nearly as many inliers as SIFT with a shorter runtime. For translation, all methods achieved around 94% inliers. Despite these numerical differences, the rotation panoramas showed similar overall alignment, with minimal differences in the image.

13 Homography Validity Analysis

I captured a third image pair by stepping sideways while photographing a nearby tree and sign against the more distant atrium. Unlike the paintings scene, these images contain considerable depth variation. Therefore, the translation produces depth dependent parallax which a single homography is not able to represent. I processed both pairs using the same SIFT, matching, RANSAC, and warping settings. The paintings pair produced 387 ratio-



Figure 17: Comparing Different Detectors and Matchers

filtered matches and 363 RANSAC inliers, an inlier ratio of 93.8%. Set 3 produced 73 matches and 37 inliers, an inlier ratio of 50.7%. Their mean inlier reprojection errors after refitting were 0.347 and 0.493 pixels, respectively.

The reference image takes up most of the panorama however, clear misalignment can be seen near the bottom of the image. Feather blending exposes this problem further, showcasing a clear doubling effect on the tree and sign in the foreground. The homography aligns some features around the balcony and lights, while the nearby tree and sign remain visibly doubled.

The low inlier error for Set 3 does not establish correct alignment throughout the scene because it measures only the retained correspondences. Even a high inlier count could reflect strong support from one dominant depth layer while objects at other depths remain misaligned. RANSAC can select a consistent subset of matches, but it cannot make a single homography represent this sort of motion.



Figure 18: Invalid Set

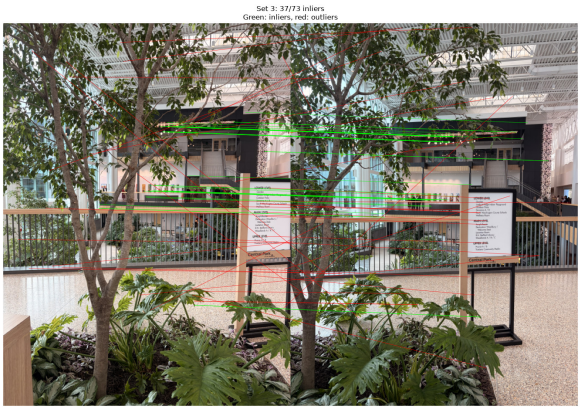


Figure 19: Invalid Inliers and Outliers



Figure 20: Invalid Panorama Comparison